CottonQTLome

User guide

A step-by-step tour of the database: how to search it, look up genes and intervals, browse and export its tables, use its genome maps, hotspots and linkage maps, and read its statistics. Each screenshot is numbered; the list under it explains every numbered item.

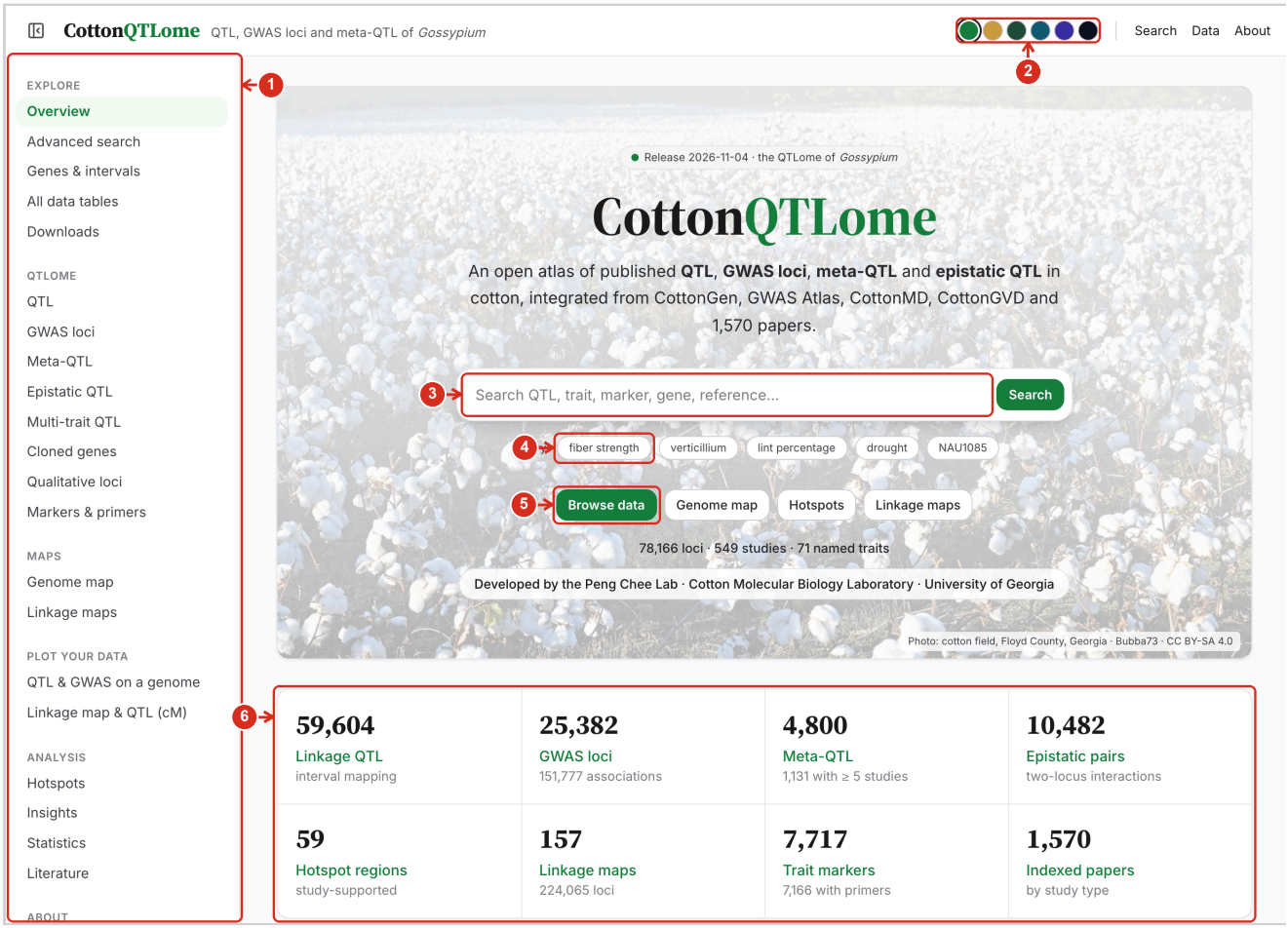
When this guide was generated the database held 59,604 linkage QTL, 25,382 GWAS loci (151,777 associations), 4,800 computed meta-QTL, 10,482 epistatic QTL pairs and 59 hotspot regions from 549 studies; 1,570 papers are indexed. Built 2026-11-04.

Web site: https://ank-man.github.io/CottonQTLome/ · Code and data: https://github.com/ank-man/CottonQTLome

1. Overview (home page)
2. Overview: traits, findings and the genome-wide view
3. Advanced search
4. All data tables
5. Working with a table
6. Meta-QTL
7. Genes and intervals: find a gene
8. Genes and intervals: narrow the genes of a region
9. Genome map: ideogram view
10. Genome map: Circos view
11. Hotspots
12. Hotspots: chromosome profile and table
13. Expression heatmap of a hotspot or meta-QTL
14. Linkage maps
15. Linkage maps: one group and the map list
16. Markers and primers
17. Statistics
18. Insights
19. Literature index
20. Multi-trait QTL
21. Cloned and validated genes
22. Downloads

1. Overview (home page)

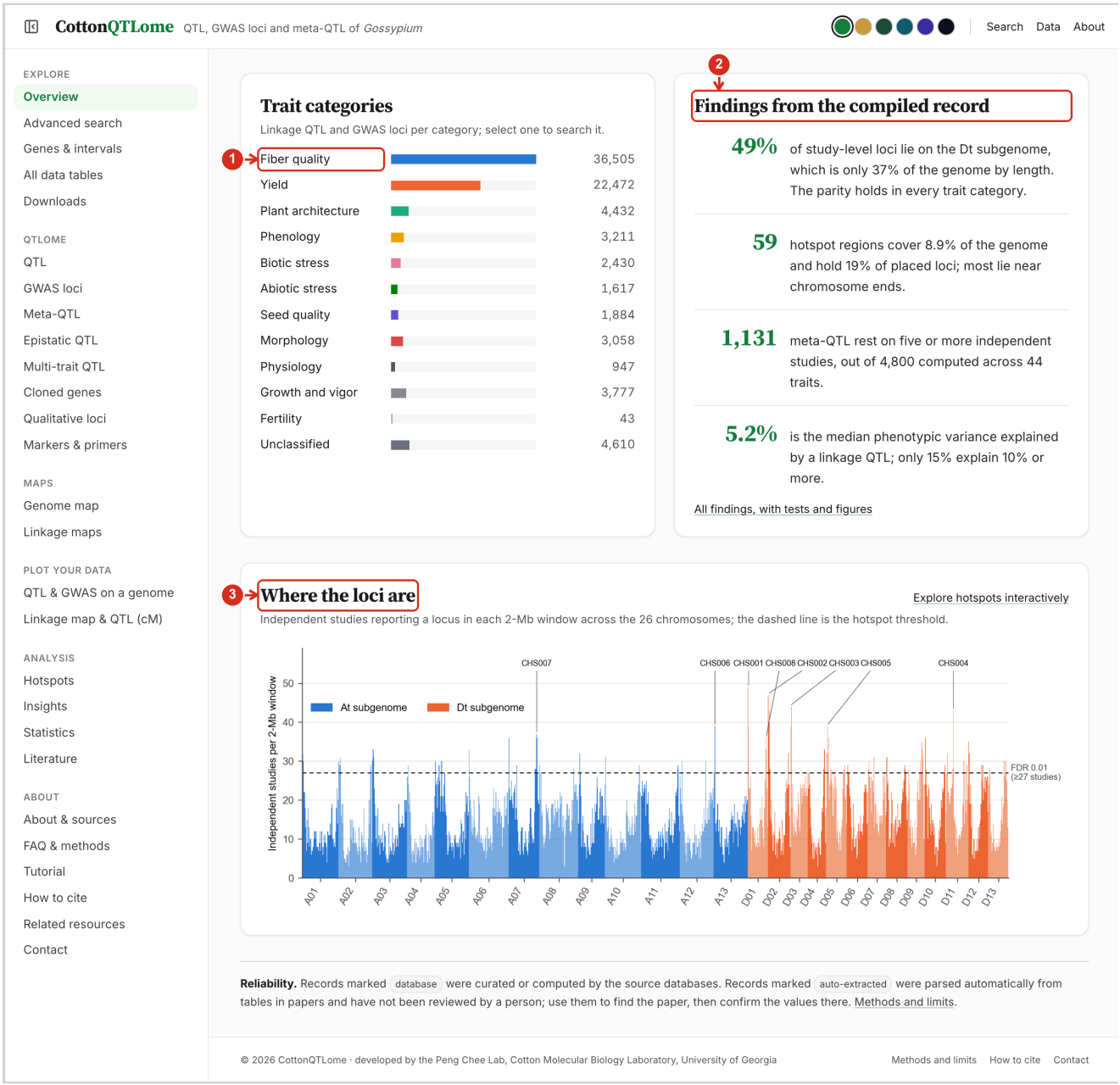
The overview page is the starting point: search everything at once, see how much the database holds, and jump to any part of it.



1	**Side menu.** Every page, grouped as Explore, QTLome (the data tables), Maps, Analysis and About. The button at the top left hides or shows it.
2	**Colour theme.** Changes the colours of the site; the choice is remembered in your browser.
3	**Search box.** Type a QTL name, trait, marker, gene or author and press Search; the Advanced search page opens with the results.
4	**Example searches.** Run a ready-made search in one click.
5	**Quick links.** Open the data tables, the Genome map, the Hotspots or the Linkage maps.
6	**Live counts.** Linkage QTL, GWAS loci, meta-QTL, epistatic pairs, hotspots, linkage maps, trait markers and papers, read from the current data. Click one to open that table.

2. Overview: traits, findings and the genome-wide view

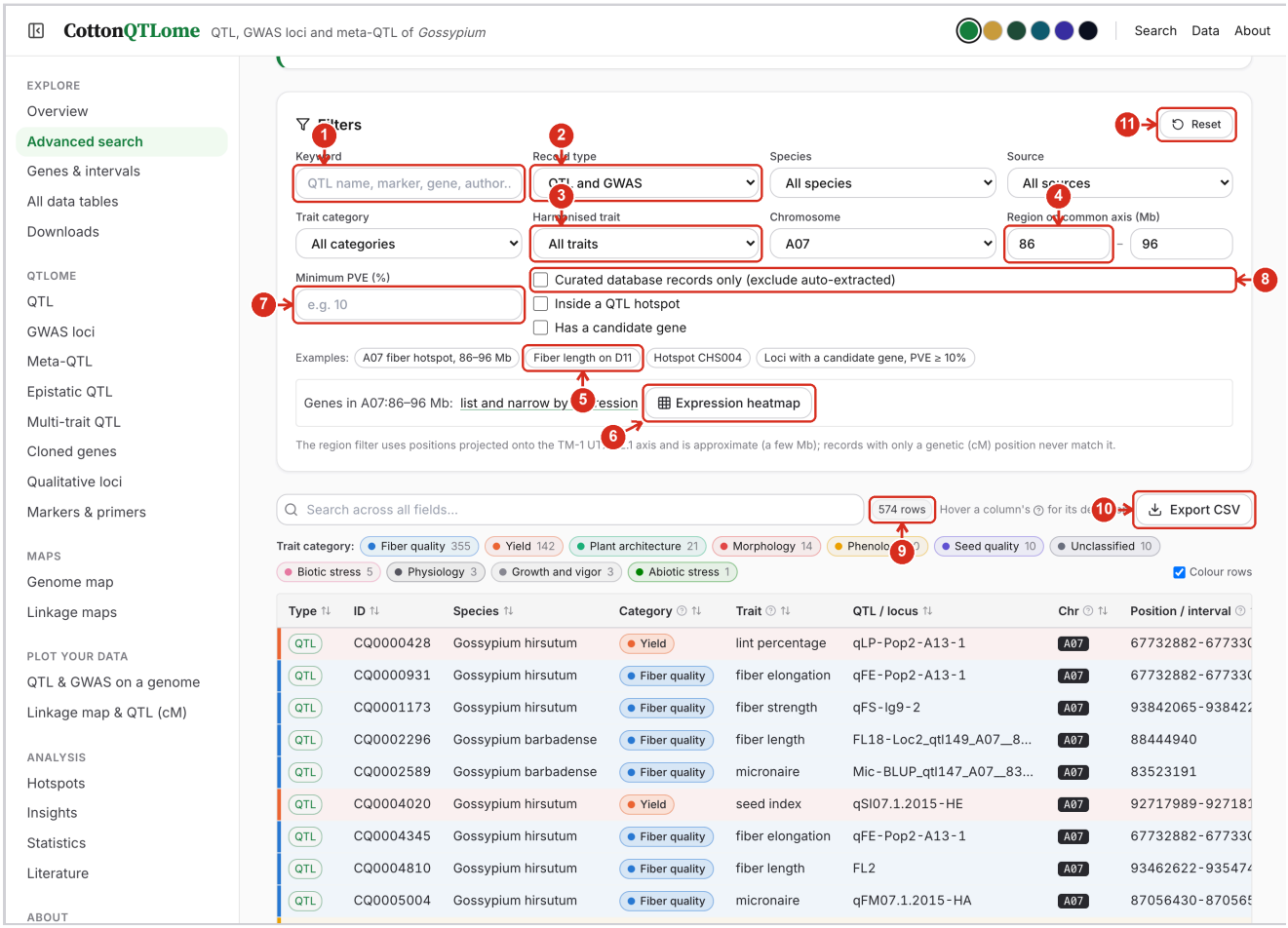
Further down the overview page are the trait categories, the main findings and a genome-wide plot of where the loci are.



1	**Trait categories.** Records per trait category. Click a category to search all of its loci.
2	**Findings.** The headline results of the analysis. “All findings, with tests and figures” opens the Insights page.
3	**Where the loci are.** Independent studies per 2-Mb window along all 26 chromosomes. Peaks above the dashed line are hotspots; “Explore hotspots interactively” opens the Hotspots page.

3. Advanced search

Advanced search filters linkage QTL and GWAS loci together. Filters apply as you set them, and the table below updates at once. Here the example “A07 fiber hotspot, 86–96 Mb” has been clicked.

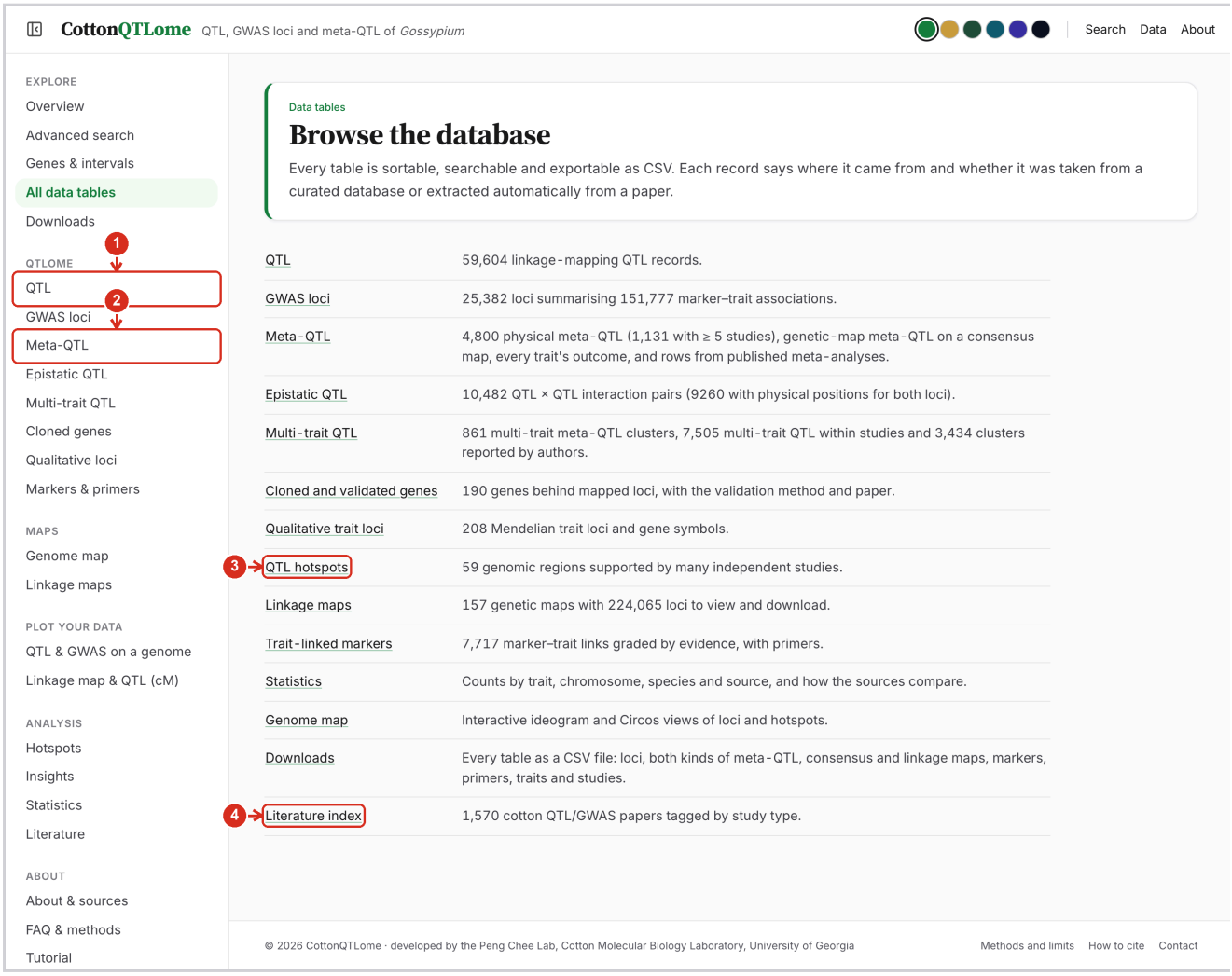


1	**Keyword.** Matches QTL name, marker, candidate gene, author and the other text fields.
2	**Record type, species and source.** Linkage QTL, GWAS loci or both; one species; one source database (CottonGen, CottonGVD, CottonMD, GWAS Atlas) or the literature.
3	**Trait category and trait.** A category such as Fiber quality, then a harmonised trait such as fiber strength. Trait names from different studies are merged into one harmonised name.
4	**Chromosome and region.** Pick a chromosome and give a start and end in Mb to list the loci in that window. Positions are projected onto TM-1 UTX v2.1, so the window is approximate; loci with only a cM position are never matched.
5	**Examples.** Ready-made searches: a hotspot region, one trait on one chromosome, the records of a hotspot, and large-effect loci with a candidate gene. Click one to fill in the filters.
6	**Genes in the region.** Appears when a chromosome and a region of up to 30 Mb are set: list the genes in the region and narrow them by organ expression, or open their expression heatmap.
7	**Minimum PVE.** Keep only loci that explain at least this percentage of phenotypic variance.
8	**Evidence filters.** Keep only records from curated databases, only loci inside a hotspot, or only loci with a candidate gene.

9	**Result count.** The number of matching rows.
10	**Export CSV.** Downloads exactly the matching rows, with every column.
11	**Reset.** Clears every filter.

4. All data tables

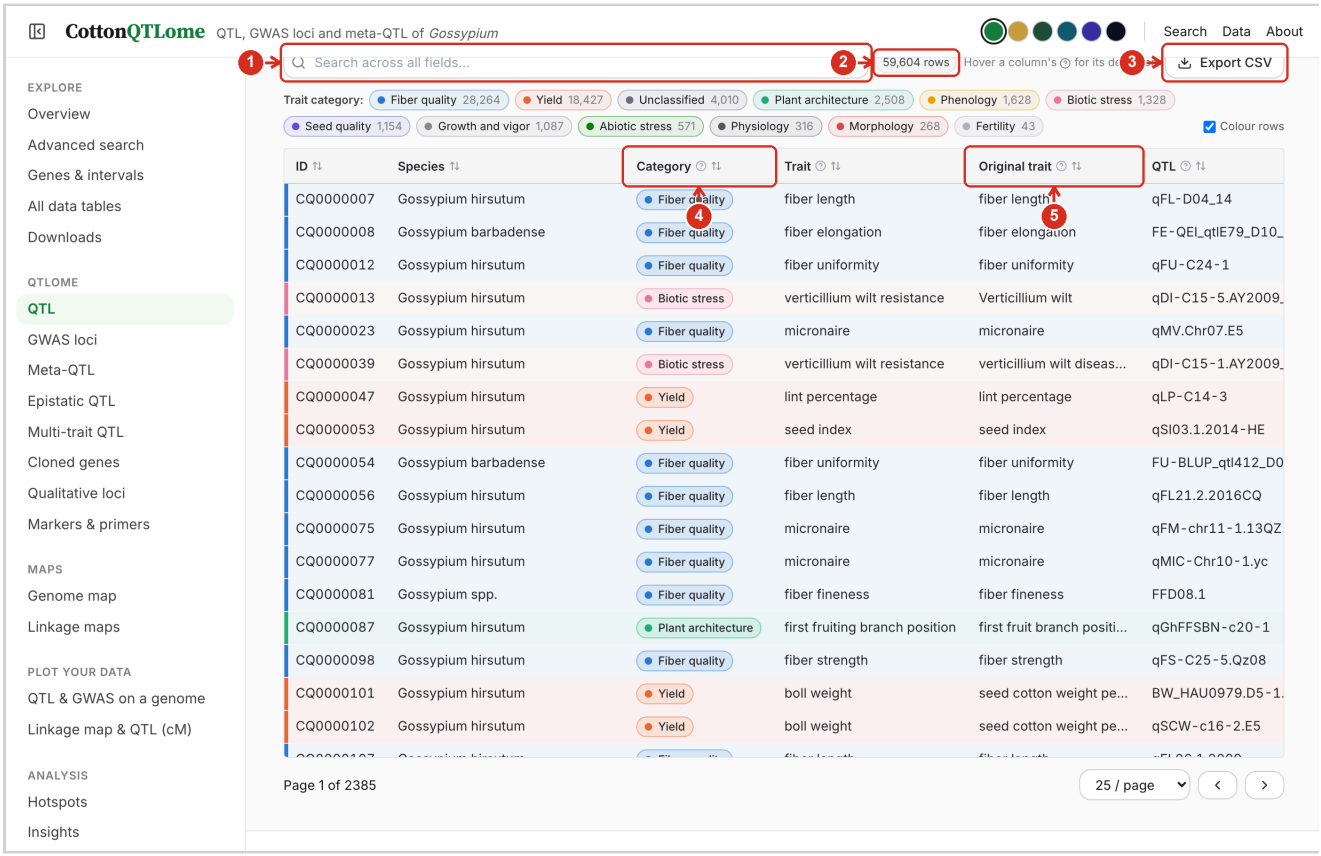
The data page lists every table with its size. Each opens as a sortable, searchable table that can be exported.



1. **QTL and GWAS loci.** Linkage-mapping QTL, and GWAS loci (each locus summarises the associations reported at one position).
2. **Meta-QTL, epistatic QTL, clusters, qualitative loci.** Consensus loci computed across studies, QTL × QTL interactions, multi-trait clusters reported by authors, and Mendelian trait loci.
3. **Hotspots, linkage maps and markers.** Study-supported hotspot regions, genetic maps with their markers, and trait-linked markers with primers.
4. **Literature index.** Every indexed cotton QTL and GWAS paper, tagged by study type.

5. Working with a table

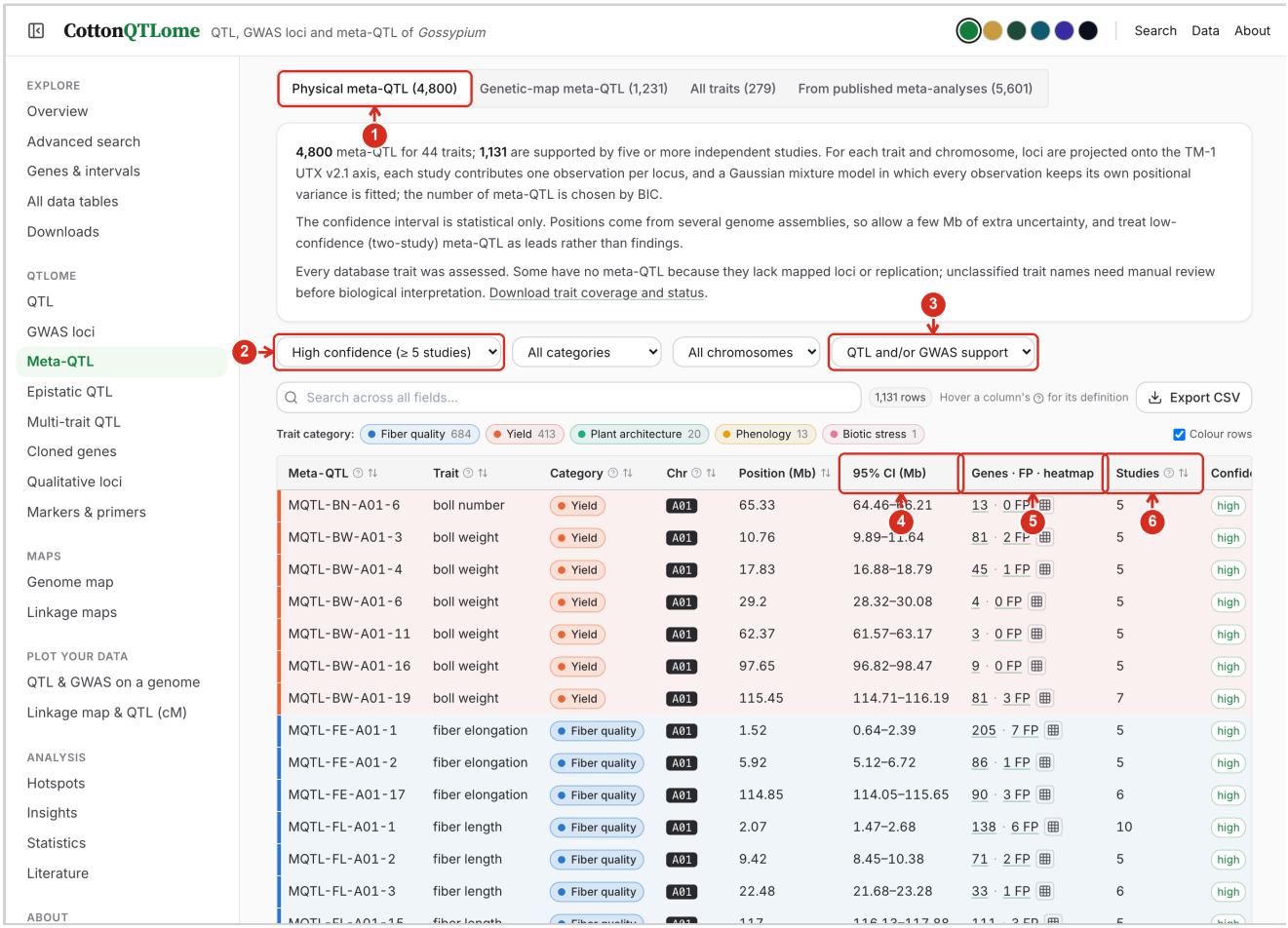
Every table works the same way, whether it is QTL, GWAS loci, meta-QTL, markers or the literature.



1	**Search the table.** Filters the rows as you type, across all columns.
2	**Row count.** The number of rows currently shown.
3	**Export CSV.** Downloads the rows currently shown.
4	**Sort and definitions.** Click a column heading to sort by it; click again to reverse. Hover the small “?” for the definition of the column.
5	**Harmonised and original trait.** Trait is the harmonised name used for searching; Original trait is the name the study used. Scroll the table sideways for position, LOD, PVE, markers, source, evidence and the reference.

6. Meta-QTL

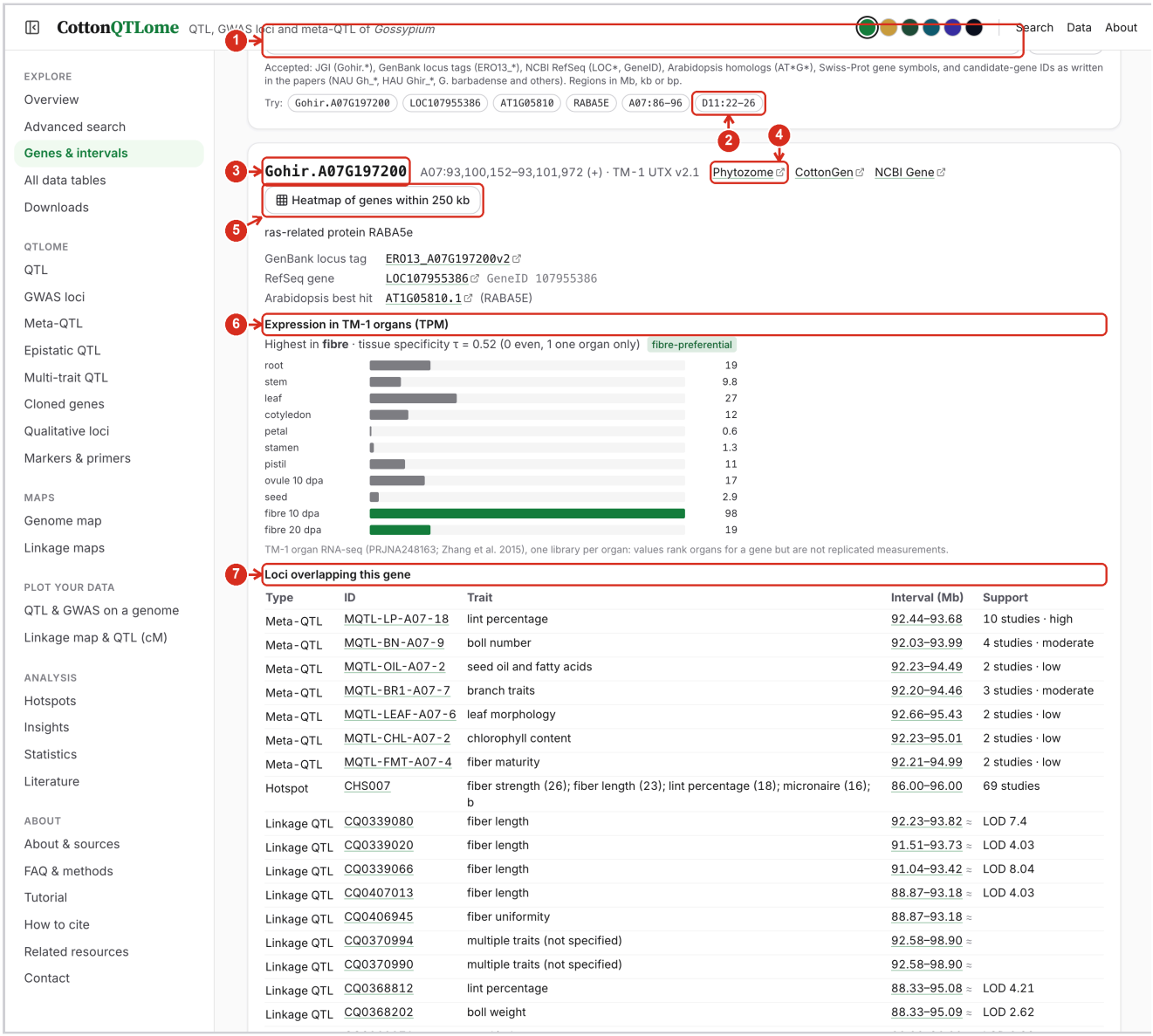
Meta-QTL are consensus loci computed from all placed QTL and GWAS loci of a trait, each supported by at least two independent studies.



1	**Four tabs.** Physical meta-QTL computed by this project (Mb), classic meta-QTL on the consensus genetic map (cM), every trait with the outcome of both meta-analyses, and rows from published meta-analyses.
2	**Confidence.** High (five or more studies), moderate (three or more) or all (two or more).
3	**Category, chromosome and support.** Limit by trait category or chromosome, or to meta-QTL supported by linkage QTL, GWAS or both.
4	**Position and interval.** Consensus position and its 95% confidence interval in Mb on TM-1 UTX v2.1.
5	**Genes, fibre genes and heatmap.** Genes in the 95% CI and how many of them are fibre-preferential (FP). Click the first number for the gene list, “FP” for the list narrowed to fibre-preferential genes, or the grid button for the expression heatmap.
6	**Studies and confidence.** How many independent studies support the meta-QTL, and the confidence level that follows from it.

7. Genes and intervals: find a gene

Look up a gene by any of its names: see its expression in 11 cotton organs and which meta-QTL, hotspots, QTL and GWAS loci cover it.



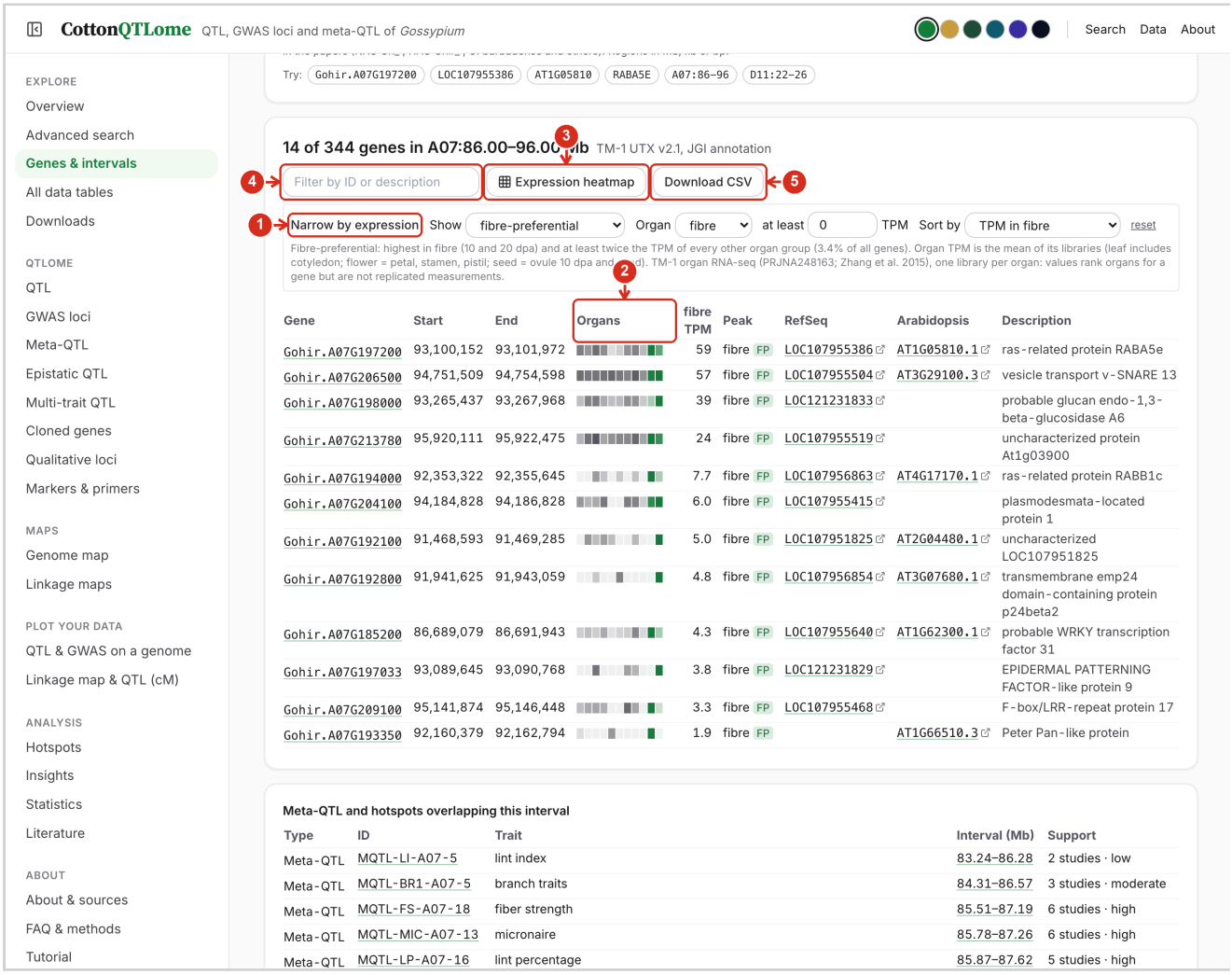
1	**Search box.** Any identifier of a TM-1 gene: JGI Gohir ID, GenBank locus tag (ERO13_…), RefSeq LOC number, NCBI GeneID, the older Gh_ IDs, or an Arabidopsis AGI code or gene symbol of its best homolog. Version and transcript suffixes are ignored. A region such as A07:90.1-91.3 lists the genes in it instead.
2	**Examples.** One click searches an example of each kind of identifier (JGI, RefSeq, Arabidopsis, gene symbol) or one of the two fibre hotspots.
3	**Gene and position.** The JGI gene ID and its position on TM-1 UTX v2.1, the assembly every map position in the database is drawn on.
4	**Genome databases.** Open the gene in Phytozome, CottonGen or NCBI Gene. The locus tag, RefSeq gene and Arabidopsis homolog below link to NCBI and TAIR.
5	**Neighbourhood heatmap.** Expression of this gene and every gene within 250 kb, as a heatmap: useful to see whether the gene stands out from its neighbours.

6 **Organ expression.** Expression (TPM) in 11 TM-1 organs, from public RNA-seq (PRJNA248163; Zhang et al. 2015); fibre in green. Above the bars: the organ group where the gene is highest, its tissue specificity τ (0 even, 1 one organ only) and, when it applies, the fibre-preferential tag (highest in fibre and at least twice every other organ group). Each organ is a single library, so the values rank organs; they do not test differences.

7 **Overlapping loci.** Meta-QTL, hotspots, QTL and GWAS loci whose interval contains the gene. Click an ID to open it in its table, or an interval to list all genes in it. ≈ marks a position placed by relative position, accurate only to a few Mb.

8. Genes and intervals: narrow the genes of a region

Every gene in an interval, and tools to cut the list down by organ expression. Here the A07 fibre hotspot (86–96 Mb) is narrowed from 344 genes to its 14 fibre-preferential genes, ranked by fibre TPM. The Genes links in the Meta-QTL and Hotspots tables open this view for their interval.



1. **Narrow by expression.** Show all genes, expressed genes (≥ 1 TPM in an organ), genes highest in a chosen organ group, or fibre-preferential genes; set a minimum TPM in that organ; and sort by position, by TPM in the organ or by tissue specificity (τ). “reset” clears these settings.

2. **Organ profile.** Eleven small cells per gene, root to fibre 20 dpa (fibre in green), shaded by $\log_2$(TPM + 1) relative to the gene’s highest organ. Hover for the values. The next columns give TPM in the chosen organ group and the peak organ; FP marks fibre-preferential genes.

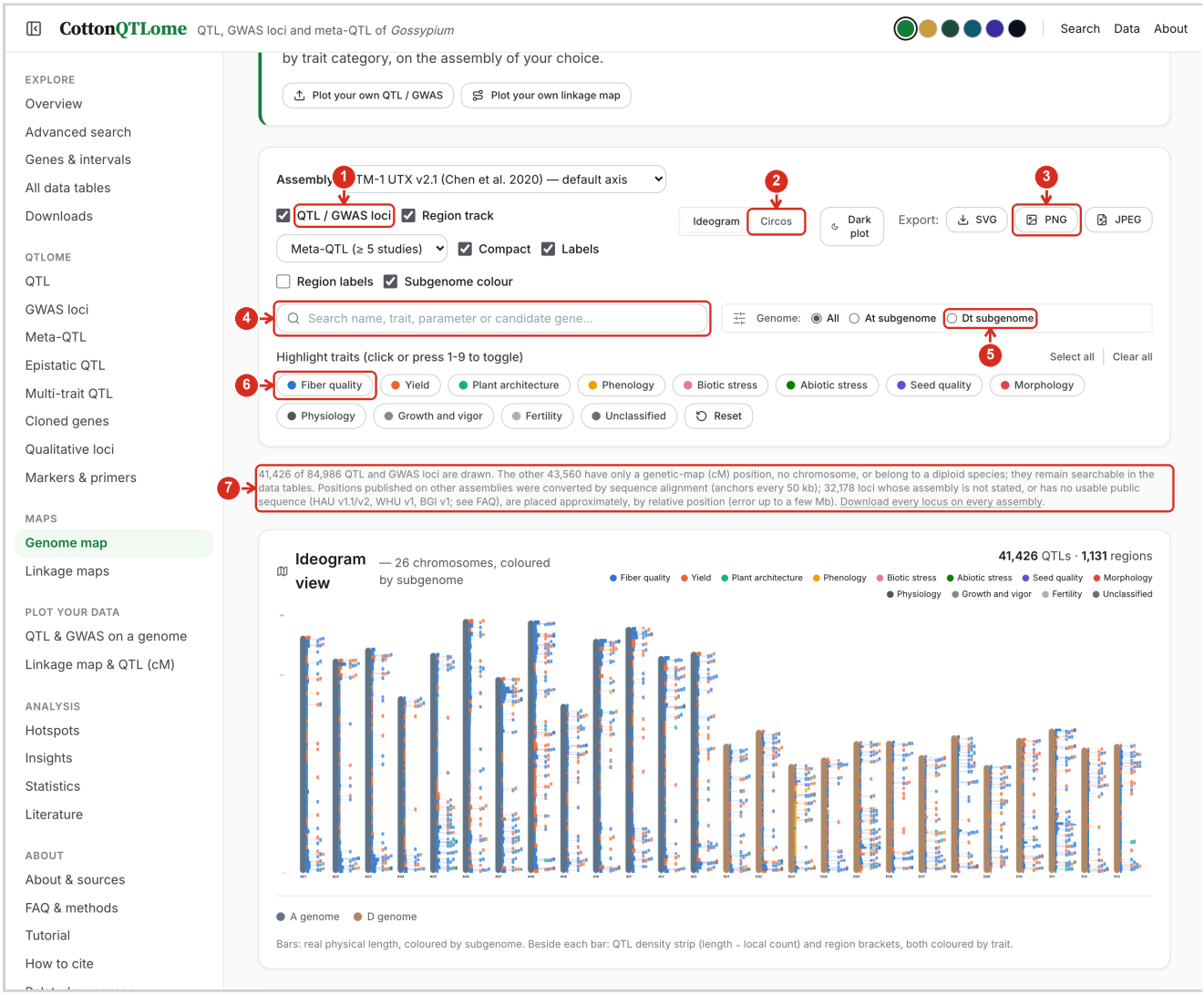
3. **Expression heatmap.** Opens the genes of the region as a heatmap of the 11 organs (see “Expression heatmap” below).

4. **Filter.** Narrow the list by any identifier or by words in the description, e.g. "kinase".

5. **Download CSV.** The genes currently listed, with all their identifiers, the description, TPM in every organ, peak organ, τ and the fibre-preferential call.

9. Genome map: ideogram view

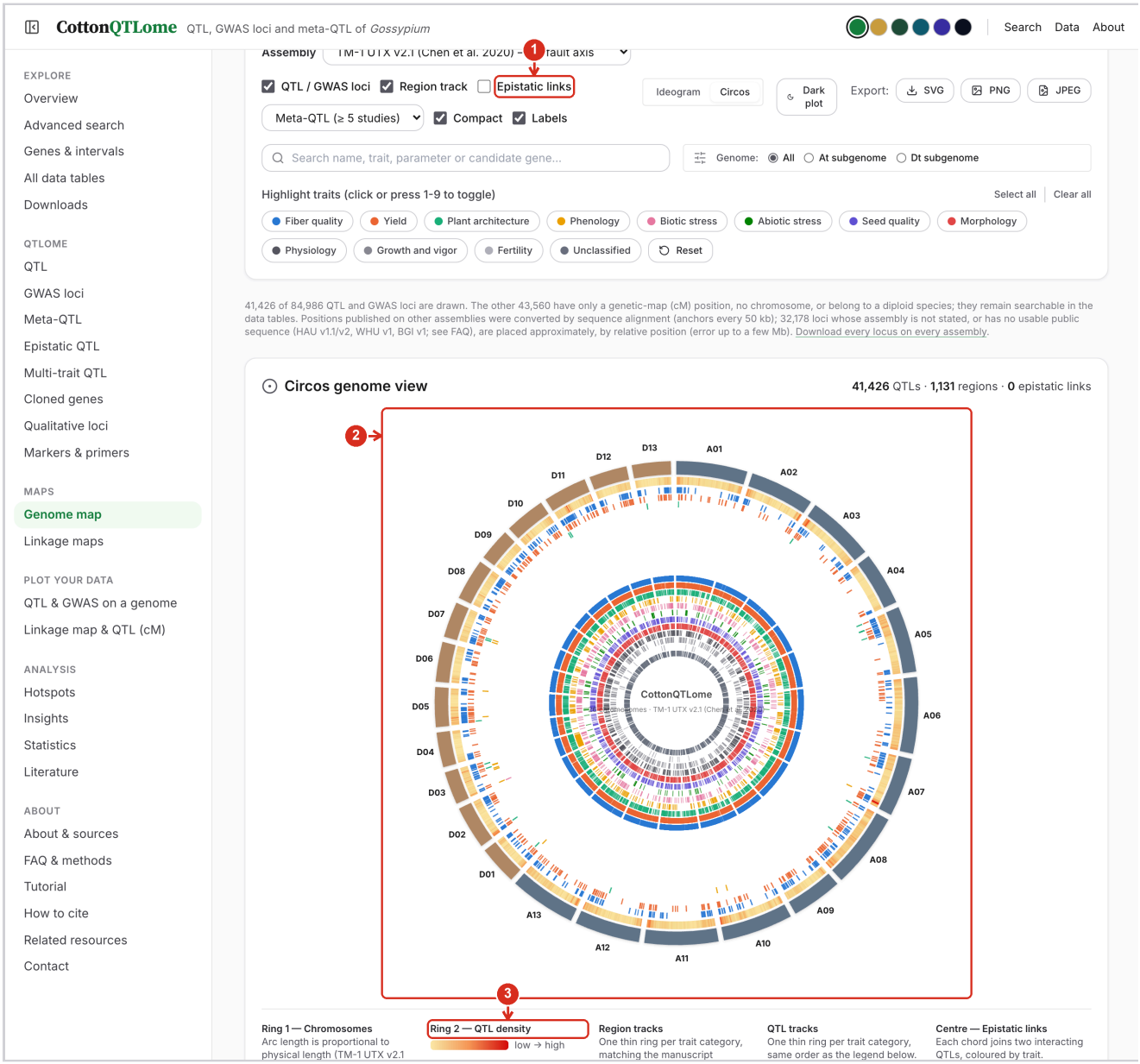
The Genome map draws loci, meta-QTL and hotspots on the 26 chromosomes of allotetraploid cotton, coloured by trait category.



1	**Layers.** Switch the loci and the region track on or off. The list chooses what the region track shows: high-confidence meta-QTL, all meta-QTL or hotspots.
2	**Choose the view.** Ideogram (chromosomes side by side) or Circos (all chromosomes in a circle). “Dark plot” draws the figure on a dark background.
3	**Export.** Save the current figure as SVG, PNG or JPEG.
4	**Search the map.** Shows only the loci whose name, trait or candidate gene matches.
5	**Subgenome.** Show all chromosomes, or only the At (A01–A13) or Dt (D01–D13) subgenome.
6	**Highlight traits.** Click one or several categories (or press 1–9) to show only those; “Reset” shows all again. The colours are those of the plot.
7	**What is drawn.** How many loci have a physical position and are drawn. The others (cM position only, or a diploid species) stay searchable in the tables.

10. Genome map: Circos view

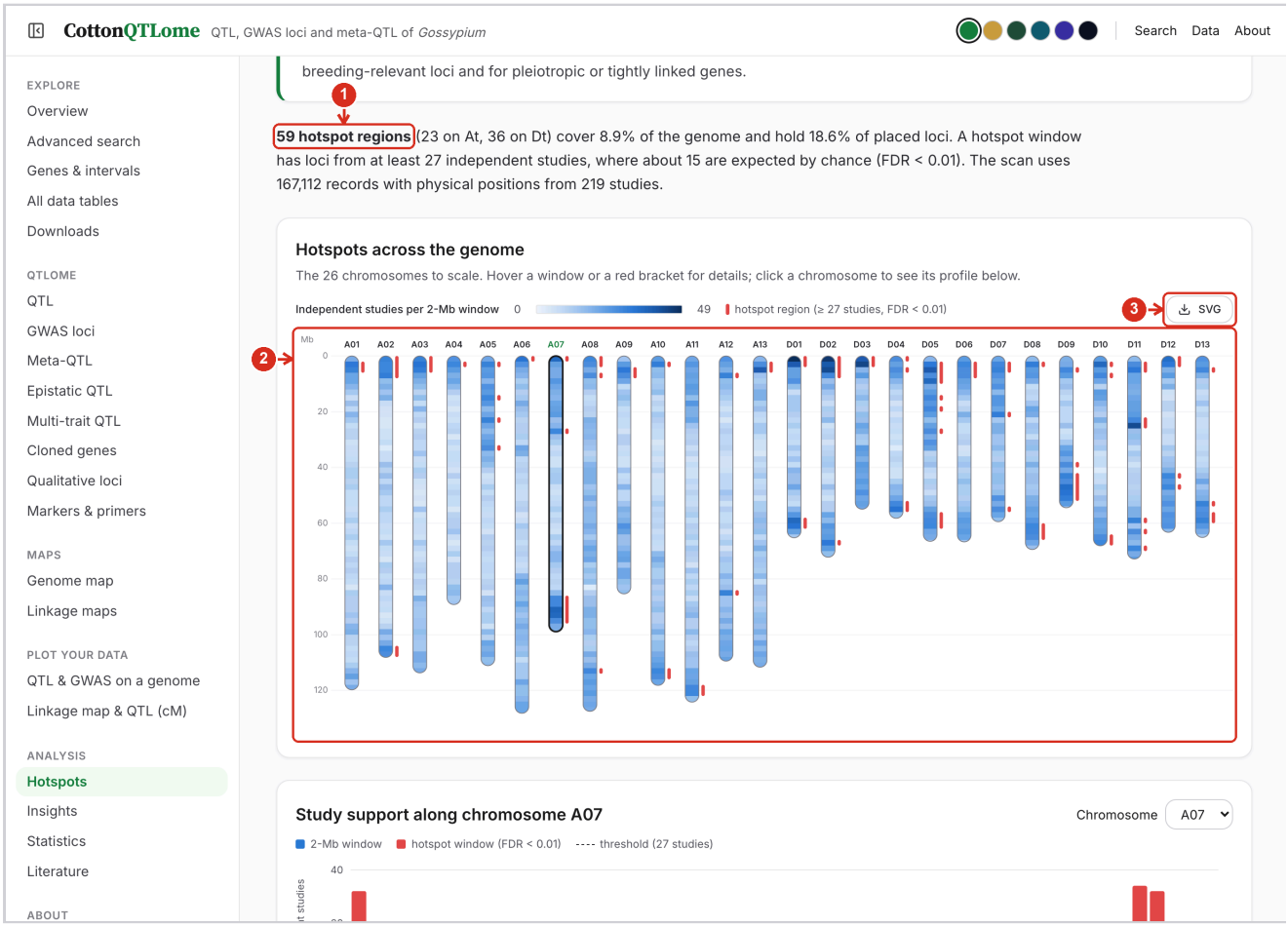
The Circos view puts all 26 chromosomes in one circle, with rings for locus density, regions and loci of each trait category.



1	**Epistatic links.** Tick to draw chords joining the two loci of each epistatic pair.
2	**The plot.** Outer ring: chromosomes, by subgenome. Next: locus density. Then one thin ring per trait category for regions and for loci. Hover any mark for its details.
3	**Key.** What each ring shows, with the colour scale for density.

11. Hotspots

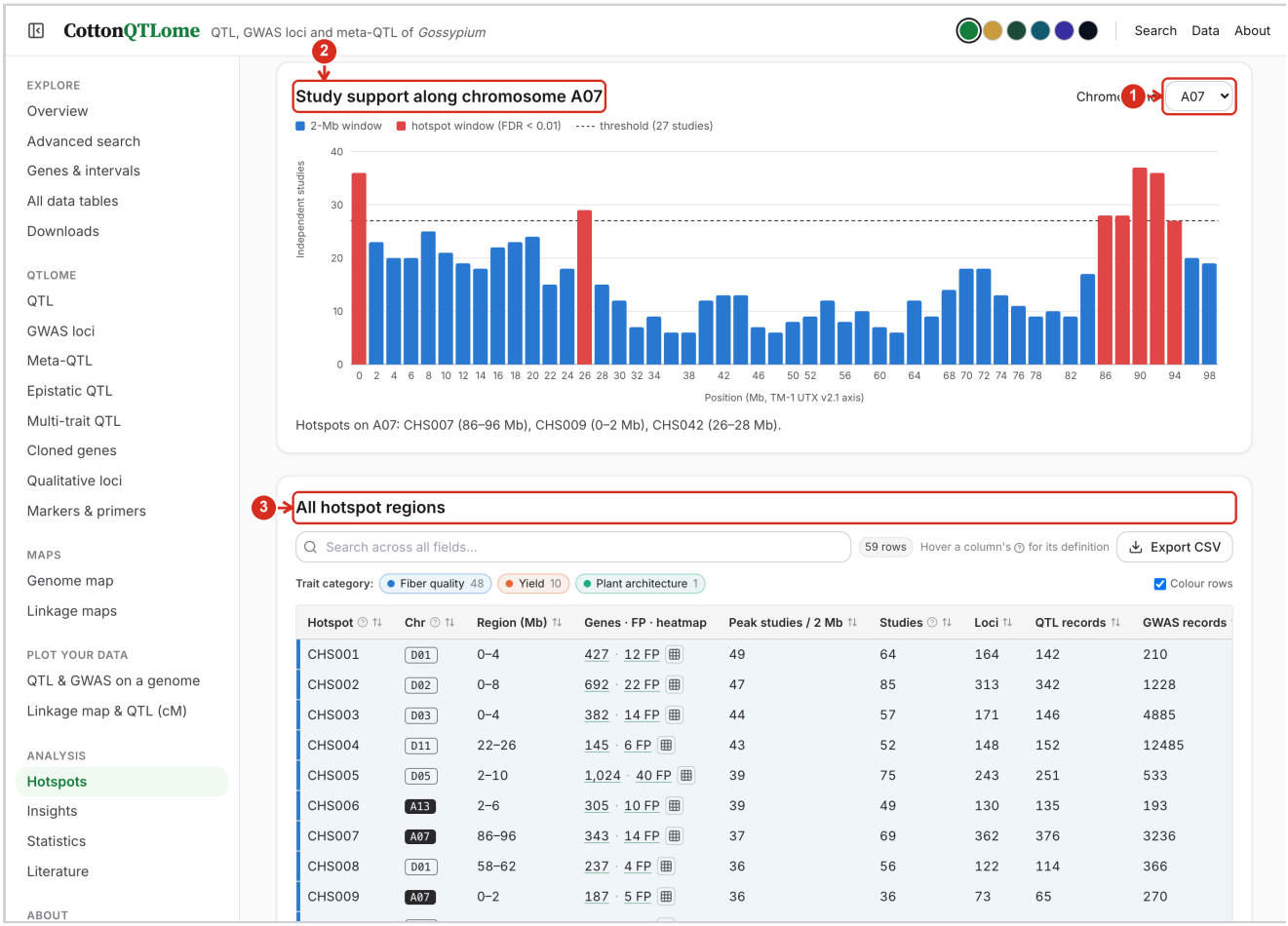
Hotspots are regions where loci from many independent studies fall together, more than expected by chance.



1	**Summary.** How many hotspots there are, how much of the genome they cover, and the study threshold used to call them.
2	**Karyotype.** All 26 chromosomes to scale. Shading shows independent studies per 2-Mb window; red brackets are hotspots. Hover for details; click a chromosome to see its profile below.
3	**Save the figure.** Downloads the karyotype as SVG.

12. Hotspots: chromosome profile and table

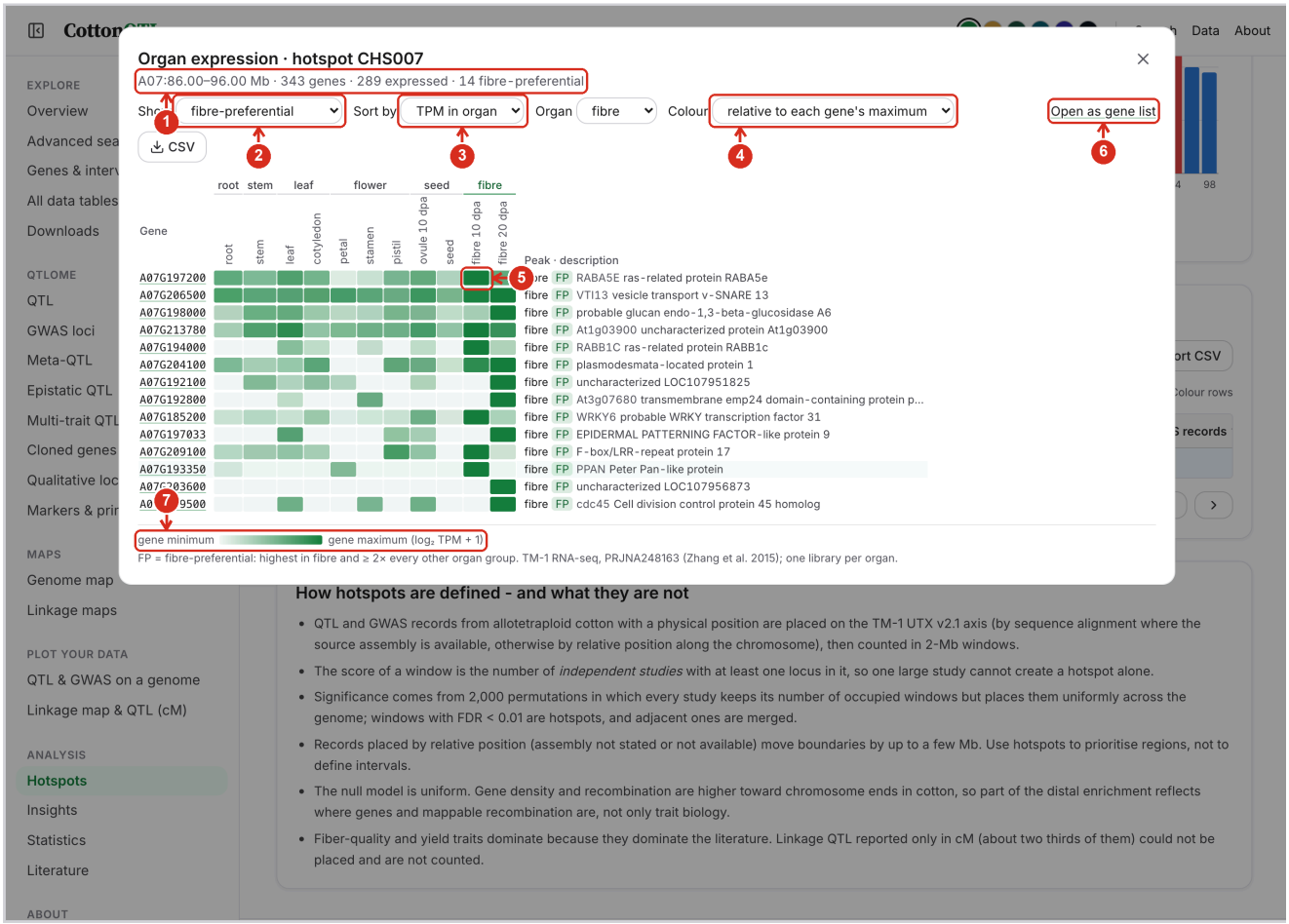
Below the karyotype, one chromosome is shown in detail, followed by the table of all hotspot regions.



1	**Chromosome.** Choose the chromosome whose study-support profile is drawn.
2	**Profile.** Independent studies per window along the chromosome; windows above the threshold form hotspots.
3	**Hotspot table.** Each hotspot with its region, number of studies and loci, top traits, example QTL and recurrent candidate genes. The “Genes · FP · heatmap” column gives the genes in the hotspot, how many are fibre-preferential, and a grid button that opens their expression heatmap.

13. Expression heatmap of a hotspot or meta-QTL

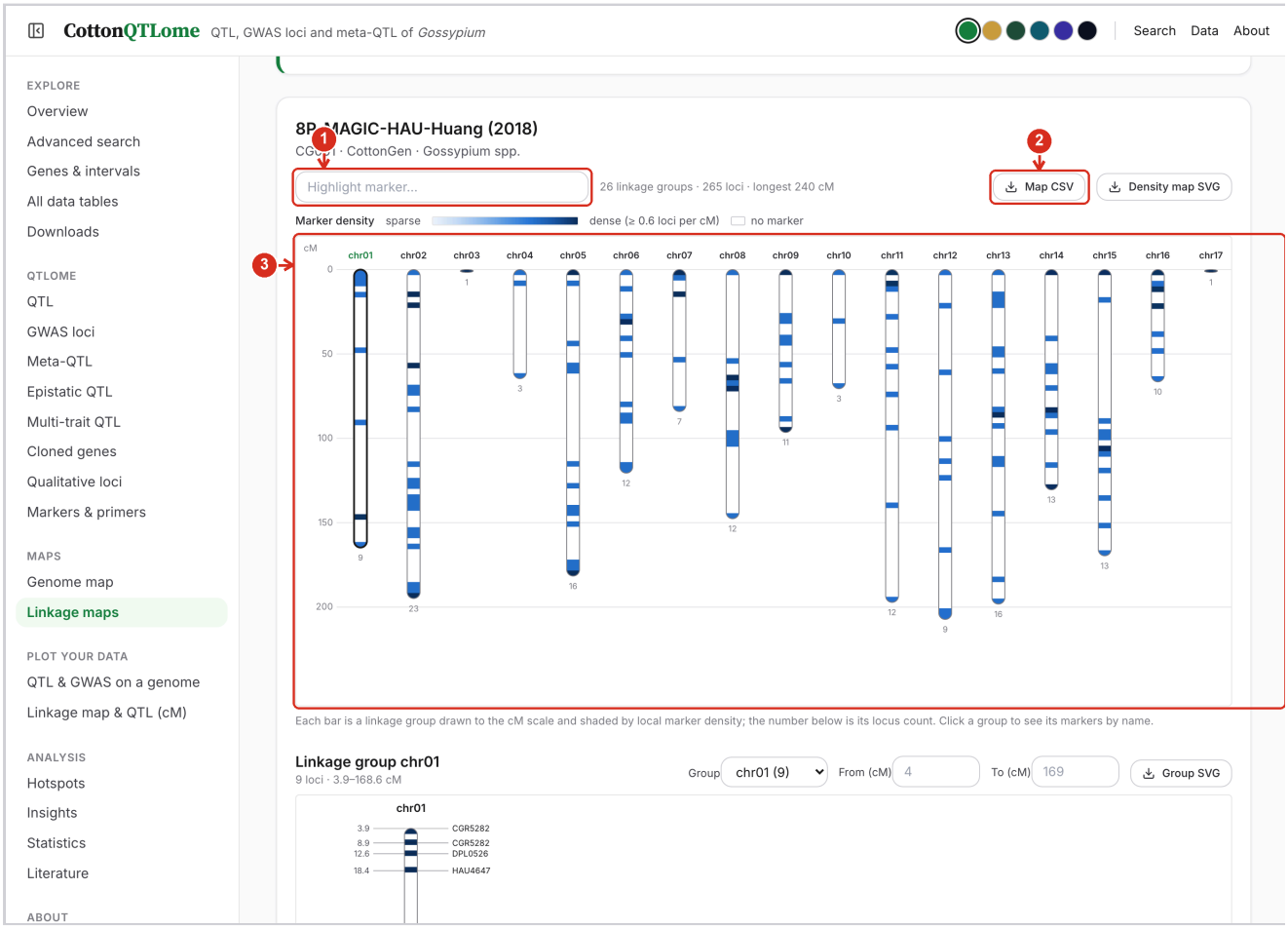
The grid button in the Genes column of the Hotspots and Meta-QTL tables, the “Expression heatmap” buttons of the gene search and the region bar of Advanced search all open this view: the genes of the interval against the 11 TM-1 organs. Here, the 14 fibre-preferential genes of hotspot CHS007 on A07.



1	**Interval summary.** Position of the interval, its genes, how many are expressed and how many are fibre-preferential.
2	**Show.** Fibre-preferential genes (the default when there are any), all expressed genes or every gene.
3	**Sort.** By TPM in a chosen organ group, by peak organ (genes grouped by the organ where they are highest), or by position.
4	**Colour scale.** Relative to each gene’s maximum shows which organ a gene prefers; the same scale for all genes shows which genes are most highly expressed.
5	**Cells.** One row per gene, one column per organ, grouped into root, stem, leaf, flower, seed and fibre. Hover a cell for its TPM; click a gene ID to open the gene.
6	**Gene list and CSV.** Open the same genes in the gene list, or download the rows shown with TPM in every organ.
7	**Key.** The colour scale ($\log_2$ TPM + 1) and the definition of fibre-preferential. Close with × or Esc.

14. Linkage maps

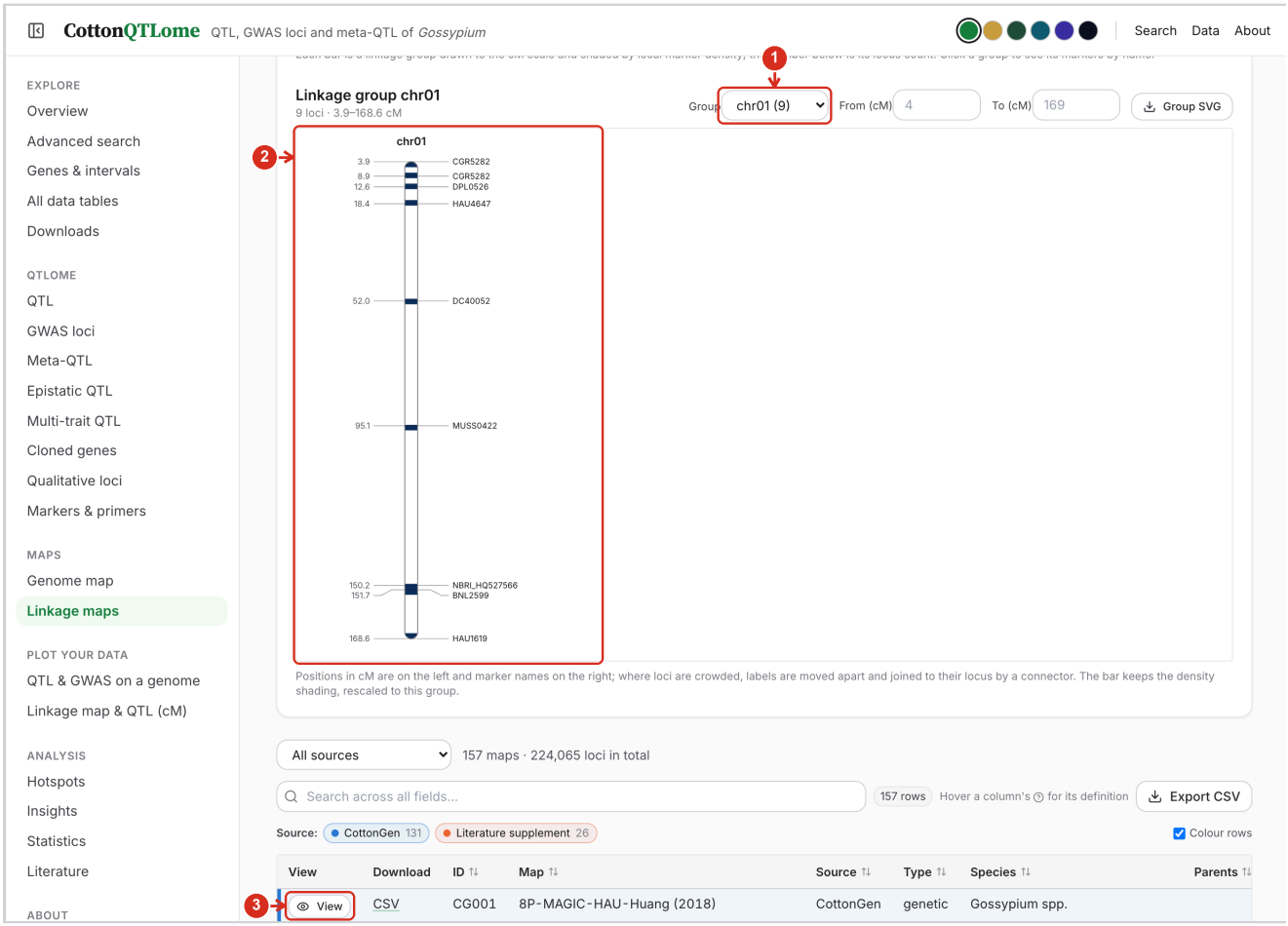
Marker orders and cM positions of cotton genetic maps, from CottonGen and from supplementary files of mapping papers.



1	**Highlight a marker.** Type a marker name to mark it on the map.
2	**Download.** Download the whole map as CSV, or the density figure as SVG.
3	**Density map.** One bar per linkage group, drawn to the cM scale and shaded by marker density; the number below is its locus count. Click a group to list its markers.

15. Linkage maps: one group and the map list

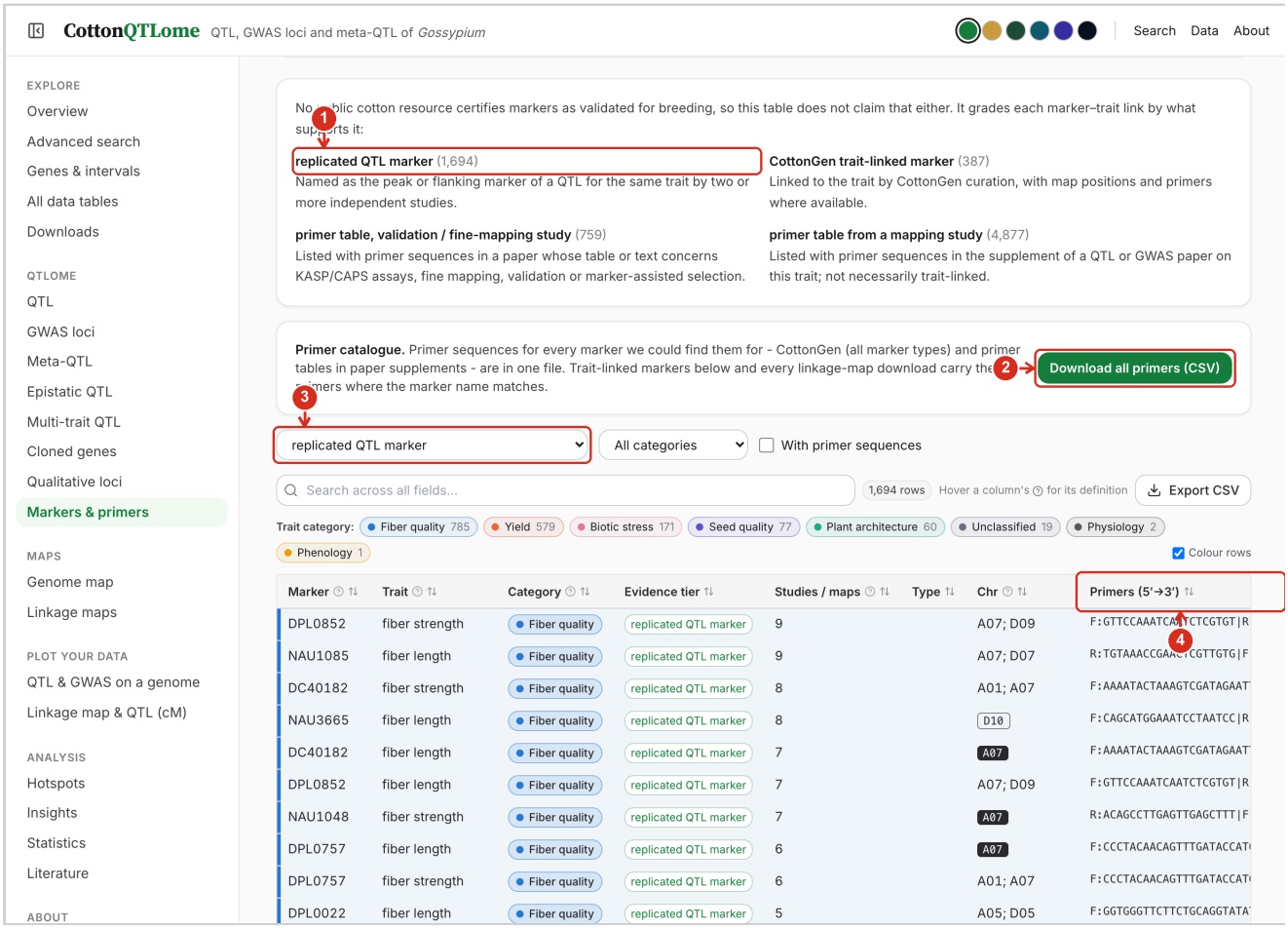
The selected linkage group is drawn with its marker names; below it every available map is listed.



1	**Group and cM window.** Choose a linkage group and the cM range to draw.
2	**Group view.** Markers in map order at their cM positions.
3	**Choose a map.** Click View on any map in the list to draw it above.

16. Markers and primers

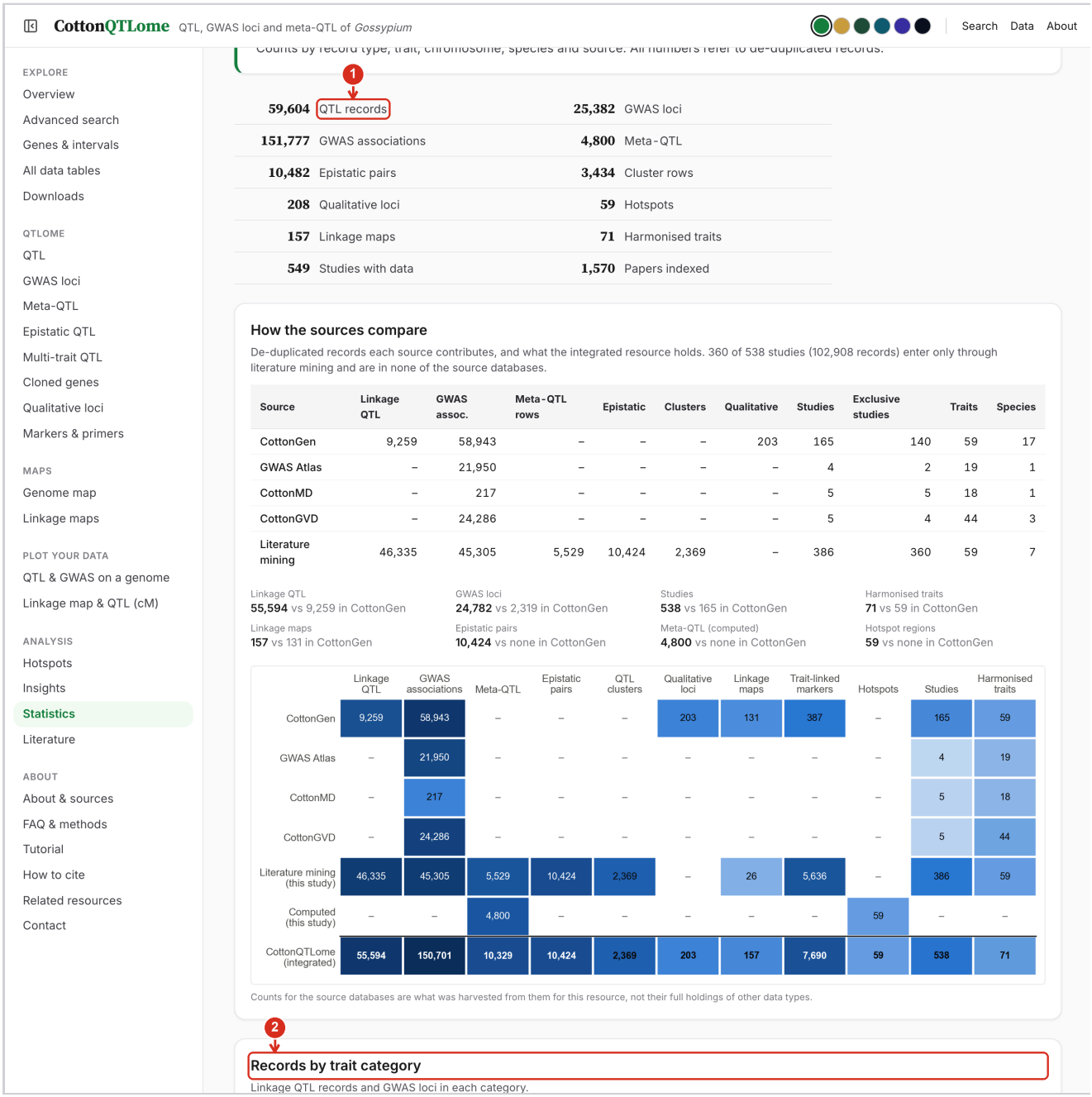
Markers linked to traits, graded by the evidence behind the link, with primer sequences where a source gives them.



1	**Evidence tiers.** What each grade means. No public resource certifies cotton markers as validated for breeding, so the grade says what supports the link.
2	**Primer catalogue.** One CSV with primers for every marker found in CottonGen and in paper supplements.
3	**Filters.** Limit by evidence tier or trait category, or to markers with primer sequences.
4	**Primers.** Forward (F) and reverse (R) primer, 5′→3′.

17. Statistics

Counts by record type, trait, chromosome, species and source. All numbers refer to de-duplicated records.

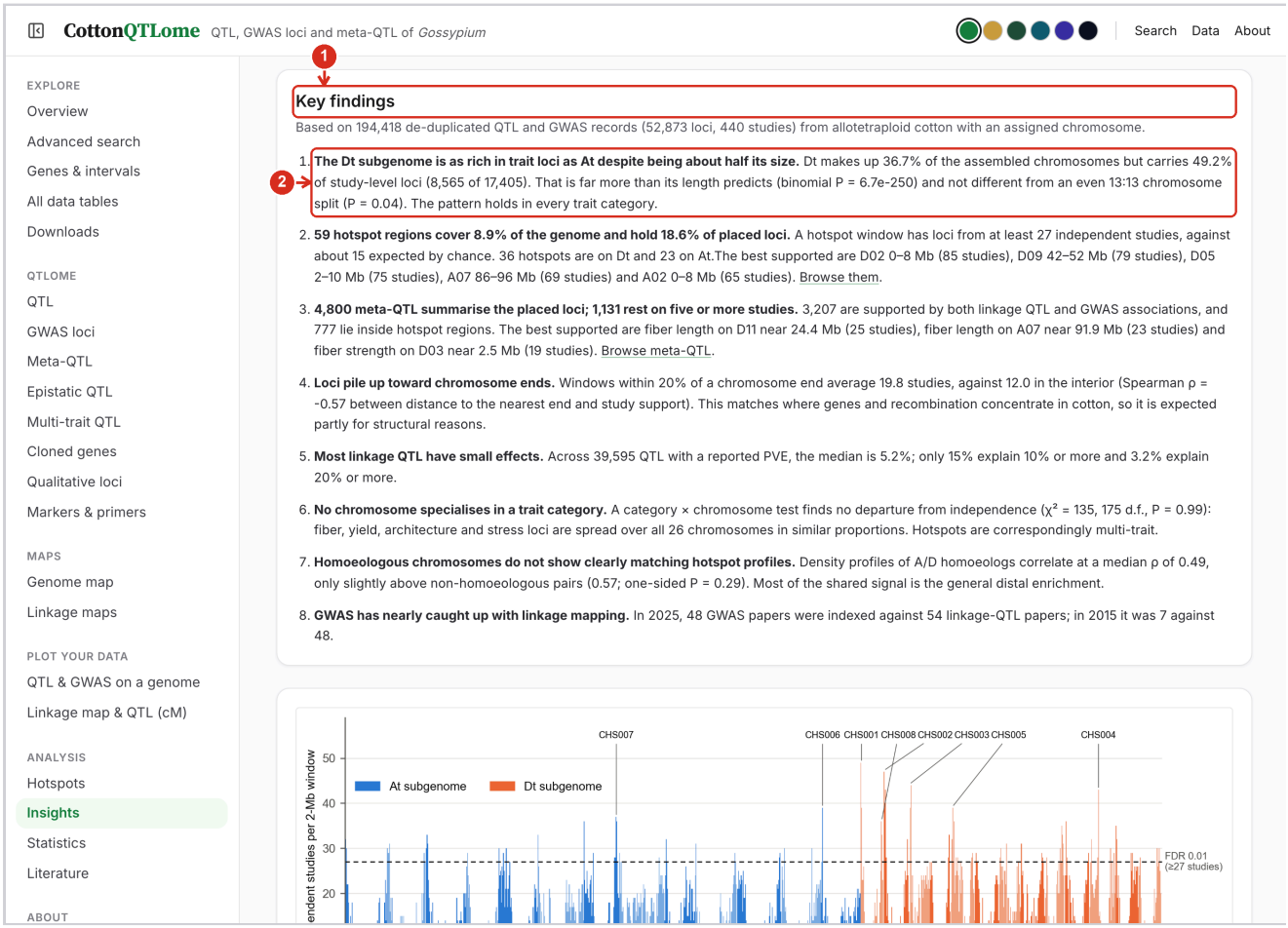


1 **Totals.** Records of every type, harmonised traits, studies with data and papers indexed.

2 **Charts.** Records by trait category, loci per Mb by chromosome, most studied traits, species, source, papers per year and the share of major-effect QTL. Hover a bar for its value.

18. Insights

What the combined record says about where trait loci sit in the cotton genome, with the statistical test behind each statement.

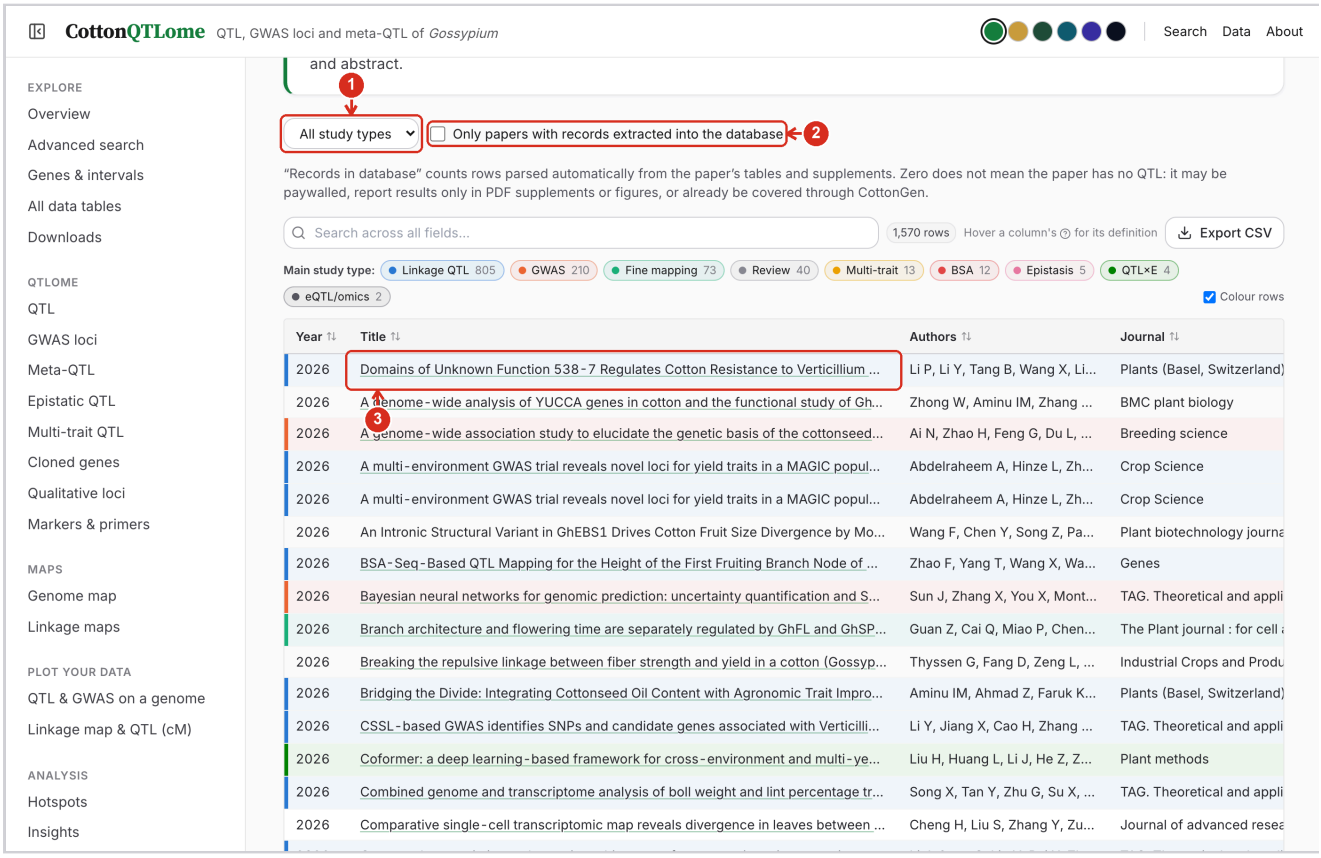


1 **Key findings.** Each finding in one sentence, then the numbers and the test. Links open the data behind it.

2 **A finding.** The figures further down the page illustrate each finding; “Read these with care” at the end lists the limits of the data.

19. Literature index

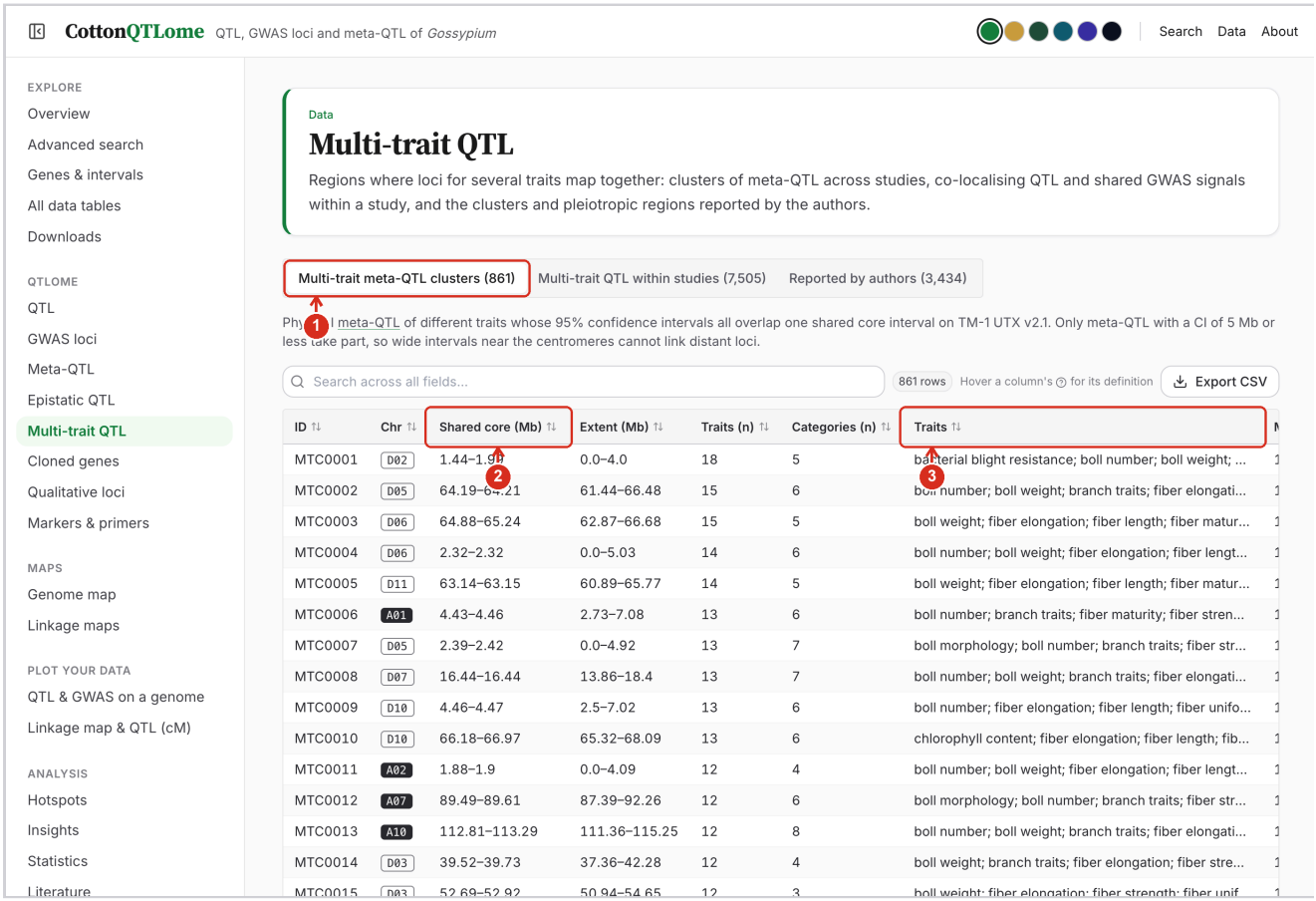
Cotton QTL, GWAS, meta-QTL and epistasis papers indexed from Europe PMC and tagged by study type.



1	**Study type.** Show only linkage QTL, GWAS, meta-QTL, epistasis or other study types.
2	**Papers with records.** Tick to show only papers whose results are in the data tables.
3	**Title.** Opens the paper. Scroll sideways for the study type and the number of records taken from it.

20. Multi-trait QTL

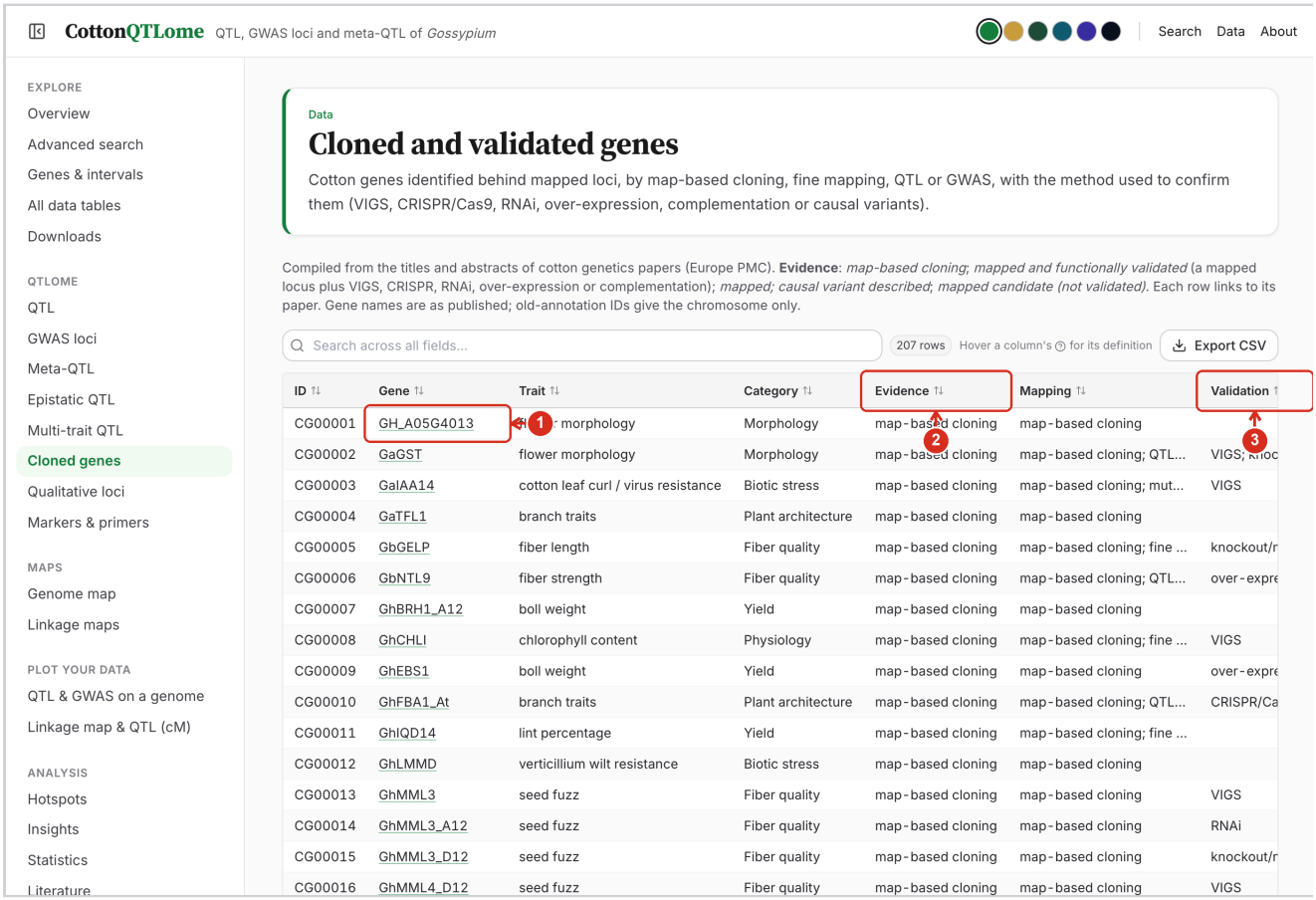
Regions where loci for several traits map together, seen three ways: clusters of meta-QTL across studies, overlapping QTL or shared GWAS signals within one study, and clusters reported by the authors.



1. **Three views.** Switch between meta-QTL clusters, multi-trait QTL within studies and the clusters reported by authors.
2. **Shared core.** The interval that all meta-QTL of a cluster overlap, on TM-1 UTX v2.1.
3. **Traits.** The traits whose meta-QTL make up the cluster; the Meta-QTL IDs column links them to the Meta-QTL page.

21. Cloned and validated genes

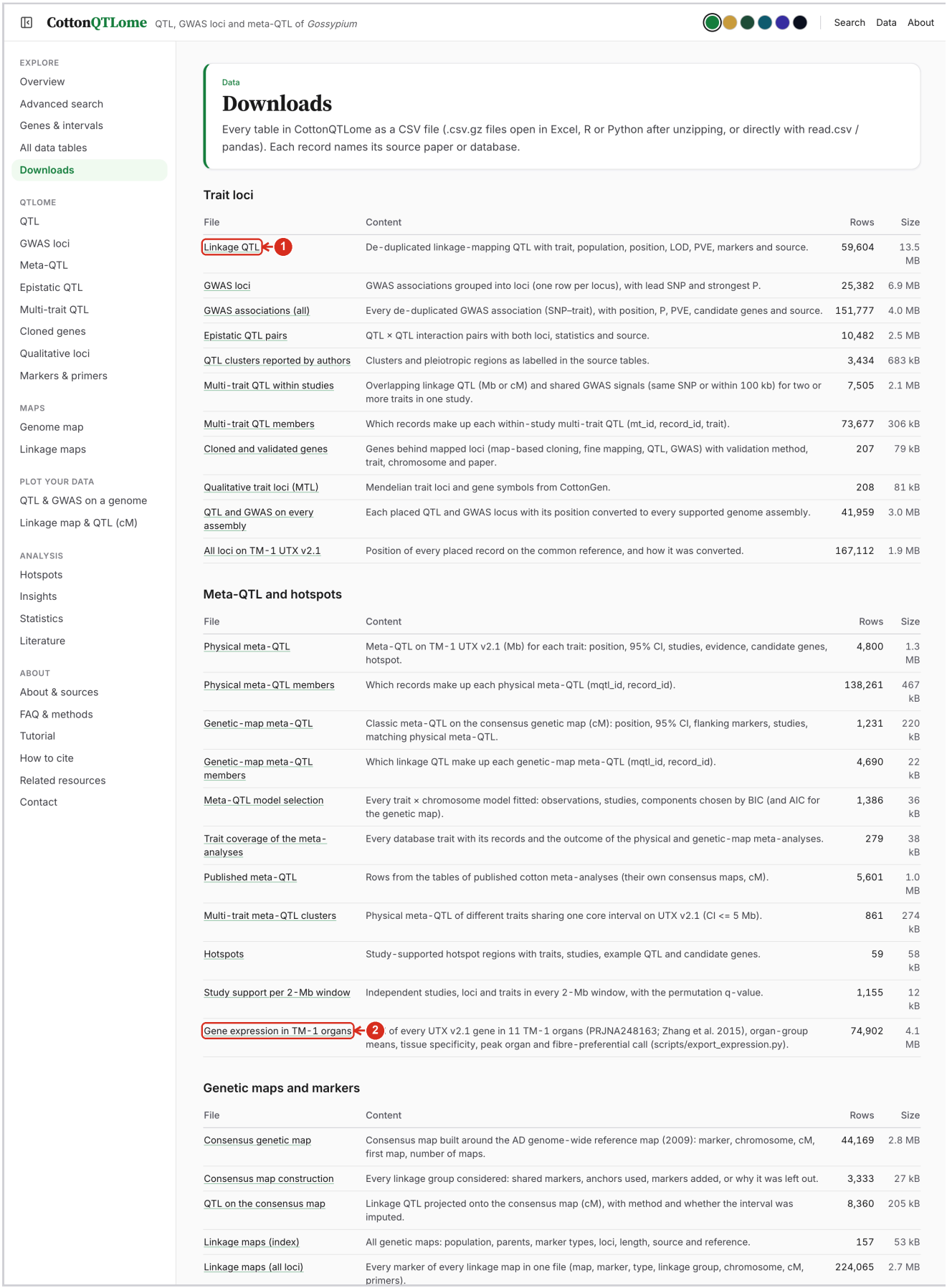
Genes identified behind mapped cotton loci, with how they were mapped and confirmed, and the paper that reports them.



1. **Gene.** Opens the gene search, which lists every QTL, GWAS locus, meta-QTL and hotspot around the gene.
2. **Evidence.** Map-based cloning; mapped and functionally validated; causal variant described; or mapped candidate.
3. **Validation.** VIGS, CRISPR/Cas9, RNAi, over-expression, complementation or allele evidence, as stated in the paper.

22. Downloads

Every table of the database as a file: loci, meta-QTL and their members, consensus and linkage maps, markers, traits, studies and validation results.



1. **Download a table.** Click a name to download the file (CSV, or gzipped CSV for the large tables).
2. **Gene expression.** TPM of every TM-1 gene in the 11 organs, organ-group means, tissue specificity, peak organ and the fibre-preferential call.